

FLUEY: Formal Logical Unified Engine for Candlestick Analysis

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Abstract

Candlestick charts are an essential and widely utilized tool in the domain of financial market analysis, particularly within the framework of technical analysis. These charts encapsulate complex price dynamics of financial instruments over discrete time intervals by encoding four fundamental values: the opening price, the closing price, the highest price, and the lowest price within each period. Traditional interpretations of candlestick formations rely heavily on qualitative descriptors such as “long upper shadow,” “small body,” “hammer,” or “shooting star,” which are inherently subjective and dependent on the visual judgment of individual traders. While these descriptors are intuitively accessible to humans and can be employed for rapid, heuristic-driven trading decisions, they lack a precise mathematical formulation, thereby limiting their reproducibility, scalability, and applicability in algorithmic trading and quantitative research contexts.

This paper introduces **FLUEY**, a fully quantitative and systematic framework designed to convert traditional candlestick pattern recognition into a deterministic and mathematically rigorous process. By defining explicit numerical rules derived from the fundamental OHLC data, FLUEY standardizes the identification of both single-candle and multi-candle patterns across varying price scales, asset classes, and market regimes. Core contributions of the framework include the derivation of normalized body ratios, precise evaluation of upper and lower shadows, threshold-based pattern classification, and a hierarchical registry system that encodes all patterns as modular, sequentially evaluable detection functions.

The proposed methodology eliminates ambiguity inherent in visual interpretation, enabling automated scanning of extensive financial time series datasets with complete reproducibility. Beyond algorithmic trading applications, FLUEY provides a foundational platform for researchers to empirically test the predictive power of candlestick patterns, refine threshold parameters, or integrate machine learning models to enhance pattern recognition accuracy. This work bridges the gap between qualitative financial intuition and rigorous quantitative modeling, making candlestick analysis both accessible and analytically precise for traders, researchers, and financial analysts.

Keywords: Candlestick Patterns, Quantitative Finance, Technical Analysis, Pattern Recognition, Algorithmic Trading, FLUEY Framework

1 Introduction

Candlestick charts are a fundamental visualization tool in financial market analysis, originating in Japan in the 18th century, where they were first used to track rice trading prices. Today, they have become a cornerstone in modern technical analysis, employed by traders and analysts globally to interpret and anticipate market movements. Each candlestick represents price activity over a discrete period, capturing the open (O), high (H), low (L), and close (C) prices in a compact and visually intuitive form. The body of the candle reflects the net price movement within the period, while the shadows, also known as wicks, illustrate intraperiod volatility and the rejection of price extremes.

Traditional candlestick analysis relies on qualitative descriptors such as “hammer,” “shooting star,” “doji,” or “marubozu,” which convey specific market psychology: a hammer suggests strong buying pressure after sellers initially push the price lower, whereas a shooting star indicates failed bullish momentum leading to potential reversals. Similarly, a doji represents market indecision, occurring when opening and closing prices are nearly identical. While these qualitative interpretations are meaningful to human traders who can quickly recognize visual patterns, they lack precise definitions suitable for algorithmic implementation, reproducibility, or large-scale quantitative analysis. Subjective interpretation introduces inconsistency across

analysts, trading platforms, and datasets, which can obscure empirical evaluation of candlestick patterns’ predictive validity.

The objective of this work is to address these limitations by introducing **FLUEY**, a fully quantitative, mathematically rigorous framework for candlestick pattern detection. FLUEY translates visual, intuitive definitions into explicit numerical criteria and threshold conditions derived from OHLC data. This approach enables deterministic identification of single-candle and multi-candle patterns, standardized across diverse price scales, asset classes, and temporal resolutions. By doing so, the framework bridges the gap between human intuition and computational precision, allowing consistent, reproducible, and scalable application of candlestick analysis to algorithmic trading, backtesting, and statistical evaluation.

This framework not only formalizes existing patterns but also provides a foundation for expanding the study of candlestick formations. Researchers and practitioners can adjust thresholds, incorporate dynamic or adaptive rules based on volatility or market conditions, and evaluate predictive performance using historical datasets. In essence, FLUEY transforms subjective chart-reading into an analytical process, accessible to both professional quantitative analysts and traders seeking a systematic methodology to integrate candlestick patterns into decision-making processes.

Key Contributions of FLUEY:

- Formalization of qualitative candlestick definitions into deterministic, numerical rules.
- Hierarchical, modular pattern detection capable of identifying single-candle to extended multi-candle formations.
- Scalable framework applicable to diverse asset classes, timeframes, and market conditions.
- Foundation for empirical evaluation and potential integration with machine learning methods.
- Bridging intuitive trading heuristics with rigorous quantitative analysis for reproducible results.

Through this introduction, the reader is prepared to understand the subsequent detailed mathematical representations, threshold-based classifications, and comprehensive pattern detection logic that constitute the FLUEY framework.

2 Mathematical Representation of Candlesticks

To transition from qualitative visual analysis to deterministic algorithmic detection, it is essential to formalize candlestick structures mathematically. Each candlestick encapsulates four fundamental price points over a given time interval: the **open price** (O), the **high price** (H), the **low price** (L), and the **close price** (C). Formally:

O = Opening price of the period, H = Highest price during the period,

L = Lowest price during the period, C = Closing price of the period

These four values fully define the price action and serve as the foundational inputs for any quantitative candlestick analysis. Each component conveys distinct market information: the opening price represents the initial market consensus, the closing price indicates the prevailing sentiment at the end of the interval, the high price reflects peak bullish pressure, and the low price captures the extreme bearish reaction.

2.1 Candle Body

The **candle body** is the most direct measure of net price movement during the interval, defined as the absolute difference between the opening and closing prices:

$$B = |C - O|$$

A large body indicates strong directional movement, whereas a small body suggests market indecision or equilibrium between buyers and sellers. By quantifying the body, we can systematically categorize candles as *large*, *small*, or near *doji*-like, a task that previously relied on subjective visual interpretation.

2.2 Price Range

The **total price range** measures the full extent of intraperiod volatility:

$$R = \max(H - L, \epsilon)$$

Here, ϵ is a small constant introduced to prevent division by zero in subsequent calculations. The range captures both the body and shadows, representing the complete spectrum of trading activity during the interval. It is critical for normalizing other metrics, ensuring consistency across price scales and assets with varying absolute prices.

2.3 Normalized Body Ratio

To standardize comparisons across assets and timeframes, we define the **normalized body ratio**:

$$\rho = \frac{B}{R}$$

This ratio expresses the proportion of the total price range occupied by the candle body. Values near 1 indicate a dominant body, whereas values near 0 suggest minimal directional movement relative to the overall volatility. This measure allows for the precise classification of candles into small, medium, or large bodies, independent of absolute price levels.

2.4 Upper and Lower Shadows

Shadows provide insight into intraperiod market rejection at extreme price points. We define:

$$U = H - \max(O, C) \quad (\text{upper shadow})$$

$$D = \min(O, C) - L \quad (\text{lower shadow})$$

The upper shadow (U) measures the distance from the top of the body to the high, representing failed bullish momentum, while the lower shadow (D) measures the distance from the bottom of the body to the low, indicating failed bearish pressure. Quantifying shadow lengths enables rigorous identification of candles like hammers, shooting stars, and doji variants, where relative shadow proportions are crucial.

2.5 Midpoint and Relative Metrics

To support multi-candle pattern analysis, we define the **midpoint** of a candle body:

$$M = \frac{O + C}{2}$$

This metric is essential for evaluating reversal patterns such as the morning star or evening star, where the closing position of subsequent candles relative to prior candle midpoints indicates momentum shifts. Relative shadow and body metrics (e.g., shadow-to-body ratios) allow formal thresholds for “long” or “short” shadows, enabling deterministic pattern recognition. For instance, a long lower shadow may be defined as $D > kB$, where k is a scalar multiplier (commonly 2.0), translating qualitative terms like “long shadow” into actionable numerical rules.

2.6 Intuition Behind the Mathematics

The mathematical formalization serves two purposes:

1. **Precision:** Ambiguous descriptors such as “small body” or “long shadow” are replaced with exact, reproducible definitions. For example, a doji occurs when $\rho \leq 0.05$, a condition derived from statistical observation of typical intraday ranges across assets.

2. **Scalability:** Normalized ratios and standardized measures ensure the framework works consistently across different price levels, asset classes, and temporal resolutions. This allows the algorithm to function

uniformly for stocks trading at \$1 versus \$1000, or on timeframes from one minute to one day, without manual recalibration.

By establishing these quantitative foundations, FLUEY transforms the inherently subjective act of candlestick pattern recognition into a rigorous, algorithmically tractable methodology. These definitions underpin subsequent sections on threshold-based classification, single- and multi-candle pattern logic, and modular detection architecture.

3 Threshold-Based Candle Classification

Having established precise mathematical representations of candlesticks, the next critical step is to classify these candles into categories that correspond to traditional trading terminology. In FLUEY, this is achieved through **threshold-based rules** applied to body size, shadow length, and relative proportions, translating qualitative human intuition into deterministic numerical criteria. These thresholds enable the framework to identify specific candles consistently across datasets and market conditions.

3.1 Doji Candles

The term *doji* originates from Japanese candlestick nomenclature and denotes a candle in which the opening and closing prices are virtually identical. Traditionally, a doji visually appears as a very thin or nearly absent body, reflecting market indecision, where neither buyers nor sellers establish dominance during the interval.

Mathematically, we define a doji using the normalized body ratio:

$$\rho = \frac{B}{R} \leq 0.05$$

Here, $B = |C - O|$ is the absolute body length, and $R = H - L$ is the total price range. The threshold 0.05 reflects the empirical observation that in most liquid assets, a candle with a body less than 5% of its total range effectively communicates indecision. This formalization allows the algorithm to detect doji candles deterministically, rather than relying on subjective visual judgment.

3.2 Small and Large Bodies

Candlestick bodies vary in size and convey the intensity of directional market pressure. In traditional analysis:

- **Small bodies** indicate indecision or weak momentum.
- **Large bodies** reflect strong market conviction in a particular direction.

FLUEY translates these qualitative descriptors into precise numerical thresholds:

$$\rho \leq 0.3 \quad (\text{small body})$$

$$\rho \geq 0.6 \quad (\text{large body})$$

These thresholds are informed by statistical analysis of historical price data and represent commonly accepted visual distinctions in traditional candlestick literature. Small bodies often appear in patterns such as spinning tops or haramis, while large bodies are characteristic of long white or long black candles, signifying robust directional movements.

3.3 Shadow Length Classification

Shadows, also referred to as wicks, capture intraperiod market reactions beyond the opening and closing prices. Their relative length indicates the degree to which extreme price levels were rejected. In conventional chart reading:

- A **long upper shadow** signals a failed attempt to push prices higher, indicating potential bearish reversal.
- A **long lower shadow** reflects a rejected bearish movement, suggesting a possible bullish reversal.

FLUEY quantifies these concepts using shadow-to-body ratios. For a shadow to be classified as “long,” the following condition is applied:

$$U > k \cdot B \quad \text{or} \quad D > k \cdot B$$

where k is a scalar multiplier, typically 2.0. This threshold ensures that only candles with significantly prominent shadows relative to their bodies are flagged as exhibiting strong rejection at extremes. The mathematical formalization removes ambiguity from traditional subjective assessments of shadow length and allows precise pattern detection for candles like hammers, shooting stars, or long-legged dojis.

3.4 Intuitive Interpretation of Thresholds

The thresholds in FLUEY serve as a bridge between human intuition and algorithmic rigor. By quantifying body size, range, and shadow proportion:

1. Traders can interpret the patterns in the context of market psychology. A large bullish candle reflects strong buying pressure, a doji conveys hesitation, and long shadows indicate failed attempts to move prices beyond a threshold.
2. The algorithm gains consistent rules applicable across diverse assets and timeframes. For example, a stock trading at \$50 or \$500 exhibits the same normalized characteristics, making detection scale-invariant.
3. Thresholds facilitate hierarchical pattern detection. Single-candle features are evaluated first, allowing multi-candle patterns to build upon deterministic candle classifications.

3.5 Extended Metrics for Multi-Candle Analysis

Thresholds also enable the quantitative evaluation of relationships between consecutive candles. For instance, a bullish engulfing pattern requires the body of the second candle to completely encompass the previous bearish candle’s body. Mathematically, this translates to conditions on the opening and closing prices relative to normalized ranges:

$$O_2 < C_1 \quad \text{and} \quad C_2 > O_1$$

Similarly, the morning star pattern depends on the second candle being “small-bodied,” satisfying $\rho_2 \leq 0.3$, and the third candle closing above the midpoint of the first. These deterministic thresholds allow FLUEY to encode multi-candle patterns unambiguously, preserving the behavioral interpretations present in classical chart reading.

3.6 Summary

Threshold-based classification provides the backbone for FLUEY’s pattern detection capabilities. By rigorously defining dojis, small and large bodies, and long shadows, the framework converts human intuition into reproducible numerical rules. This classification not only facilitates accurate identification of single-candle patterns but also forms the foundation for more complex multi-candle structures, enabling hierarchical and deterministic detection across financial datasets of arbitrary size and complexity.

4 Single-Candle Pattern Definitions

Single-candle patterns form the foundational elements of candlestick analysis. Each pattern represents a discrete market sentiment captured during a single trading interval. In FLUEY, these patterns are precisely defined using the quantitative measures described in Sections 2 and 3, including body size, shadow lengths, and normalized ratios. This section provides a comprehensive explanation of each single-candle pattern, linking its traditional name, visual interpretation, market meaning, and exact mathematical definition.

4.1 Hammer

The hammer is one of the most recognizable bullish reversal candles. The term “hammer” arises from its visual resemblance to a hammer: a small body near the top of the candle, with a long lower shadow extending downward, often two to three times the body length. Traditionally, hammers appear after a downtrend, signaling that sellers pushed prices lower but buyers regained control, closing near the high.

Mathematically, in FLUEY, a hammer is defined as:

- Bullish close: $C > O$
- Long lower shadow: $D > 2B$

where $B = |C - O|$ is the body and $D = \min(O, C) - L$ is the lower shadow. The multiplier 2 ensures that only significantly prominent lower shadows are considered, differentiating hammers from ordinary candles with minor lower wicks. The small body near the top combined with a long lower shadow encodes the market psychology of rejected downward pressure.

4.2 Hanging Man

The hanging man is conceptually similar to the hammer but appears after an uptrend, indicating potential bearish reversal. Its name reflects the ominous visual cue—a small body at the top with a long lower shadow. Traditionally, traders interpret the hanging man as a warning that buyers are losing control and sellers are beginning to exert influence.

FLUEY defines a hanging man as:

- Bearish close: $C < O$
- Long lower shadow: $D > 2B$

This formalization isolates the critical features distinguishing hanging men from hammers: context (prior trend) and closing direction.

4.3 Inverted Hammer

The inverted hammer is another bullish reversal candle. Visually, it has a small body near the bottom, with a long upper shadow, resembling an upside-down hammer. The long upper shadow signifies that buyers attempted to push prices higher but were initially rejected, yet the bullish close indicates eventual upward momentum.

Mathematically:

- Bullish close: $C > O$
- Long upper shadow: $U > 2B$
- Short lower shadow: $D < B$

This combination captures the subtle psychology of initial resistance followed by buying strength.

4.4 Shooting Star

The shooting star is the bearish counterpart to the inverted hammer. Occurring after an uptrend, it signals a potential reversal as buyers fail to maintain higher prices. Its appearance—a small body near the low with a long upper shadow—is reminiscent of a falling star, giving it its name.

Mathematically, FLUEY defines:

- Bearish close: $C < O$
- Long upper shadow: $U > 2B$

The long upper shadow emphasizes rejected bullish attempts, while the small body indicates that buyers could not sustain momentum.

4.5 Dragonfly Doji

The dragonfly doji has virtually no body, a long lower shadow, and negligible upper shadow, resembling a “T” shape. This pattern indicates strong intraperiod selling that was completely rejected by buyers. The name “dragonfly” reflects the slender, elongated appearance of the candle.

FLUEY defines:

- Body ratio: $\rho \leq 0.05$
- Long lower shadow: $D > 2 \cdot U$

The asymmetry between upper and lower shadows quantifies the dramatic rejection of lower prices.

4.6 Gravestone Doji

The gravestone doji is the mirror image of the dragonfly doji, with a long upper shadow and negligible body and lower shadow. This candle signals strong intraperiod buying that was rejected by sellers, potentially indicating a bearish reversal. The name derives from its resemblance to a gravestone: a small base with a towering vertical line above.

FLUEY defines:

- Body ratio: $\rho \leq 0.05$
- Long upper shadow: $U > 2 \cdot D$

This definition precisely encodes the rejection dynamics captured visually by traders.

4.7 Spinning Top

A spinning top has a small body with relatively balanced upper and lower shadows. It signifies market indecision, where neither buyers nor sellers dominate. The term “spinning top” comes from the toy-like appearance of the candle, with a thin central body and small symmetrical wicks.

FLUEY criteria:

$$\rho \leq 0.3$$

This ensures the body is small relative to the range, while the shadows are neither extremely long nor dominant, preserving the visual balance.

4.8 Long White Candle

The long white candle represents strong bullish momentum. It has a large body, closes near the high, and minimal upper and lower shadows. Its name comes from traditional charting, where bullish candles were often depicted in white or light colors.

Mathematically:

- Bullish close: $C > O$
- Large body: $\rho \geq 0.6$

This formalization isolates candles that reflect strong buying pressure, aiding the identification of potential trend continuation.

4.9 Long Black Candle

The long black candle is the bearish analog of the long white candle. It reflects strong selling pressure and often follows or continues downtrends. Its name derives from traditional black or dark coloring for bearish candles.

FLUEY defines:

- Bearish close: $C < O$
- Large body: $\rho \geq 0.6$

This captures strong downward conviction unambiguously.

4.10 Bullish and Bearish Marubozu

Marubozu candles have virtually no shadows, indicating uninterrupted directional movement. “Marubozu” means “shaved” in Japanese, reflecting the lack of wicks.

FLUEY defines:

- Bullish Marubozu: $C > O, U < \epsilon, D < \epsilon$
- Bearish Marubozu: $C < O, U < \epsilon, D < \epsilon$

These definitions ensure that only candles with negligible rejection at either end are classified as marubozu, reflecting extreme market dominance.

4.11 Bullish and Bearish Belt Hold

Belt hold candles have large bodies and small shadows, typically opening at the extreme of the range and closing strongly in one direction. They suggest strong directional conviction at the start of a move.

Mathematically:

- Bullish Belt Hold: $C > O, |O - L| < \epsilon, U < 0.2 \cdot B$
- Bearish Belt Hold: $C < O, |O - H| < \epsilon, D < 0.2 \cdot B$

This formalization encodes the traditional visual identification into precise numerical rules.

4.12 Long-Legged Doji

The long-legged doji features a near-zero body with long upper and lower shadows, reflecting extreme indecision. Traders visually identify it as a candle with a thin central body and symmetrical long wicks extending above and below.

FLUEY defines:

$$\rho \leq 0.05, \quad U > 2B, \quad D > 2B$$

These criteria capture the market’s simultaneous testing of both extremes, encoded quantitatively.

4.13 Rickshaw Man

The rickshaw man is a variant of the long-legged doji, where the body is near the midpoint of the candle range, resembling a rickshaw balanced between two extremes. It reflects market equilibrium with high volatility.

FLUEY defines:

$$\rho \leq 0.05, \quad |\text{midpoint} - O| < 0.1 \cdot (H - L)$$

This captures the visual centrality and small body proportion.

4.14 High Wave Candle

High wave candles exhibit small bodies with long upper and lower shadows. They convey heightened uncertainty and increased volatility. Visual identification involves spotting the small central body flanked by elongated wicks.

FLUEY defines:

$$\rho \leq 0.3, \quad U > B, \quad D > B$$

This quantitative encoding isolates candles with pronounced volatility but small net movement, reflecting indecision in turbulent markets.

4.15 Summary

Through FLUEY’s mathematical formalization, each single-candle pattern is precisely defined in terms of body size, shadow proportions, and relative positions. This approach translates traditional visual interpretations into deterministic rules, bridging human intuition and algorithmic reproducibility. By establishing these definitions, FLUEY forms the foundation for hierarchical pattern recognition, enabling reliable detection of multi-candle patterns in subsequent sections.

5 Multi-Candle Pattern Logic

While single-candle patterns capture isolated market sentiment, true trend and reversal detection often requires analyzing sequences of multiple candles. Multi-candle patterns encode interactions between consecutive trading periods, revealing more complex market psychology that single candles alone cannot convey. In FLUEY, these patterns are defined as deterministic functions of sequential single-candle features, enabling both visual and mathematical identification.

5.1 Two-Candle Patterns

Two-candle patterns involve interactions between a base candle and a confirming candle. The critical logic in FLUEY considers:

1. Trend context: Whether the sequence occurs during an uptrend, downtrend, or sideways market.
2. Relative positioning: Comparing opens, closes, highs, and lows of consecutive candles.
3. Shadow and body analysis: Examining the proportions of each candle to classify engulfing, harami, and counterattack patterns.

5.1.1 Bullish Engulfing

The bullish engulfing pattern occurs when a small bearish candle is immediately followed by a larger bullish candle that “engulfs” the first. This indicates a potential reversal from downtrend to uptrend.

- **Visual:** Small red candle followed by large green candle overlapping previous body completely.
- **Mathematical definition in FLUEY:**

$$C_2 > O_2, \quad O_2 < C_1, \quad C_2 > O_1$$

where the second candle fully encompasses the first body.

- **Reasoning:** The engulfing reflects the abrupt shift in control from sellers to buyers, quantifiable through body overlap.

5.1.2 Bearish Engulfing

The bearish engulfing pattern is the inverse: a small bullish candle is followed by a larger bearish candle, signaling a potential reversal from uptrend to downtrend.

- **Visual:** Small green candle followed by large red candle engulfing the previous body.
- **Mathematical definition:**

$$C_2 < O_2, \quad O_2 > C_1, \quad C_2 < O_1$$

5.1.3 Bullish and Bearish Harami

Harami patterns feature a large candle followed by a small candle contained entirely within the first body, signaling indecision or trend reversal.

- **Bullish Harami:**

$$C_1 < O_1, \quad C_2 > O_2, \quad O_2 > C_1, \quad C_2 < O_1$$

- **Bearish Harami:**

$$C_1 > O_1, \quad C_2 < O_2, \quad O_2 < C_1, \quad C_2 > O_1$$

- **Explanation:** The first candle establishes momentum, and the small second candle within the range reflects hesitation, quantifiable by body containment.

5.1.4 Harami Cross

The harami cross occurs when the second candle in a harami pattern is a doji. This emphasizes indecision and amplifies reversal probability.

- **FLUEY definition:**

$$\text{Body ratio of candle 2} \leq 0.05, \quad \text{contained within candle 1}$$

- **Reasoning:** The doji highlights complete market uncertainty at the intersection of prior trend forces.

5.1.5 Counterattack Patterns

Counterattack patterns occur when the second candle opens opposite the first candle’s close and closes at or near the first candle’s close, signaling potential trend reversal.

- **Bullish Counterattack:**

$$C_1 < O_1, \quad O_2 < C_1, \quad C_2 \approx C_1$$

- **Bearish Counterattack:**

$$C_1 > O_1, \quad O_2 > C_1, \quad C_2 \approx C_1$$

- **Reasoning:** The counterattack demonstrates immediate equilibrium restoration after initial momentum, mathematically represented by proximity of closes.

5.1.6 Matching High, Kicking, Separating Lines, Tasuki Gap Up

Other two-candle formations incorporate more complex relationships involving trend continuation or reversal:

- **Matching High:** Two consecutive candles with equal highs, signaling potential resistance testing.
- **Kicking:** Abrupt gap between two opposite-direction candles, reflecting strong sentiment shift.
- **Separating Lines:** Second candle opens at the previous close, continuing trend.
- **Tasuki Gap Up:** A bullish continuation pattern involving a gap and a confirming third candle.

Each is formalized in FLUEY using candle opens, closes, highs, and lows, quantified thresholds for gap sizes, and relative body sizes.

5.2 Three-Candle Patterns

Three-candle patterns capture more nuanced market psychology, encoding sequences of acceleration, reversal, or exhaustion.

5.2.1 Morning Star

The morning star consists of a long bearish candle, followed by a small indecisive candle (spinning top or doji), and a long bullish candle. It signals bullish reversal after a downtrend.

$$C_1 < O_1, \quad \rho_2 \leq 0.3, \quad C_3 > O_3, \quad O_3 > \frac{O_1 + C_1}{2}$$

Explanation: The first candle establishes selling pressure, the second indicates indecision, and the third confirms a shift to bullish sentiment.

5.2.2 Evening Star

The evening star is the bearish counterpart:

$$C_1 > O_1, \quad \rho_2 \leq 0.3, \quad C_3 < O_3, \quad O_3 < \frac{O_1 + C_1}{2}$$

It encodes a peak formation followed by reversal.

5.2.3 Three White Soldiers and Three Black Crows

These are trend continuation patterns:

- **Three White Soldiers:** Three consecutive bullish candles with increasing closes, small or absent shadows.
- **Three Black Crows:** Three consecutive bearish candles.

Mathematical definitions:

$$C_i > C_{i-1}, \quad \rho_i \geq 0.6 \quad (i = 2, 3)$$

These formulas capture the persistent directional strength in quantitative terms.

5.2.4 Other Three-Candle Patterns

Patterns such as Three Inside Up/Down, Three Outside Up/Down, Tri Star, Unique Three River Bottom, Upside Gap Two Crows, and Deliberation are similarly defined in FLUEY with:

- Sequential body ratios
- Relative position of opens/closes/highs/lows
- Trend direction preconditions

Each pattern encodes specific market sentiment trajectories over three trading intervals.

5.3 Four- and Five-Candle Patterns

These larger patterns, including Concealing Baby Swallow, Three Line Strike, Rising Three Methods Extended, and Ladder Bottom, represent highly structured sequences often used to confirm major trend continuation or reversal. FLUEY defines them by:

- Hierarchical analysis of single- and two-candle patterns
- Cumulative body and shadow ratios
- Inter-candle gap and overlap conditions

These definitions allow algorithmic recognition of complex trends that traditional chartists detect visually.

5.4 Mathematical Rationale

The FLUEY framework translates the heuristic rules of candlestick analysis into deterministic mathematical inequalities. Each multi-candle pattern definition:

1. Reduces subjectivity in visual identification
2. Quantifies shadow and body interactions
3. Encodes trend context and reversal likelihood

This approach allows both professional analysts and automated systems to detect complex market behavior consistently.

5.5 Summary

Multi-candle patterns extend the logic of single-candle signals by incorporating temporal sequencing, relative positioning, and cumulative market psychology. FLUEY’s formalization ensures that these complex structures can be reliably and reproducibly identified in real market data, forming the basis for predictive analytics and algorithmic trading strategies.

6 Full FLUEY Algorithm Implementation and Code Walkthrough

The FLUEY framework not only defines candlestick patterns mathematically, but also implements them in a complete algorithmic system capable of scanning large time-series datasets efficiently. This section provides an in-depth walkthrough of the FLUEY code, highlighting the logic behind each function, the design rationale, and how the system integrates single- and multi-candle pattern detection.

6.1 Configuration and Thresholds

At the foundation of FLUEY lies a configuration block defining thresholds and constants that are universally applied across all pattern definitions:

- **DOJI_BODY = 0.05**: A candle with a body less than 5% of its total range is considered a doji, representing indecision.
- **LARGE_BODY = 0.6**: Bodies occupying more than 60% of the candle’s range are classified as large, signifying strong directional movement.
- **SMALL_BODY = 0.3**: Bodies below 30% of the range are considered small, often indicating minor movement or indecision.
- **LONG_SHADOW_MULT = 2.0**: A shadow twice the size of the body is considered long, highlighting extreme price rejection.
- **GAP_TOL = 0.001**: A tolerance value for gap detection between consecutive candles.
- **EPS = 1e-9**: A small constant used to avoid division by zero in normalized calculations.

These thresholds allow FLUEY to translate qualitative visual heuristics into precise numerical rules.

6.2 Helper Functions

FLUEY utilizes modular helper functions to compute candle properties. Each function abstracts a single metric:

- **body(o,c)**: Returns absolute difference between open and close. Quantifies net price movement.
- **rng(h,l)**: Returns candle range, i.e., high minus low. Includes **EPS** to avoid zero division.
- **ratio(o,c,h,l)**: Normalized body ratio $\rho = \frac{|C-O|}{H-L}$. Standardizes body size across price scales.
- **upper(o,c,h)** and **lower(o,c,l)**: Compute upper and lower shadow lengths.
- **bullish(o,c)** and **bearish(o,c)**: Boolean functions to identify direction of candle.
- **is_doji(o,c,h,l)**, **is_large(o,c,h,l)**, **is_small(o,c,h,l)**: Categorize candle bodies by normalized ratios.
- **midpoint(o,c)**: Computes body midpoint for use in patterns like Morning Star.
- **gap_up(prev_h, curr_l)** and **gap_down(prev_l, curr_h)**: Identify gaps between consecutive candles.
- **long_upper_shadow(o,c,h)**, **long_lower_shadow(o,c,l)**: Boolean checks for extreme shadow lengths relative to body.

Each helper function allows subsequent pattern definitions to focus on logic rather than repetitive calculations, improving clarity and maintainability.

6.3 Single-Candle Pattern Functions

Each one-candle pattern is implemented as a Python function:

- **hammer(d,i)**: Identifies bullish hammers by checking for a bullish close ($C > O$) and a lower shadow at least twice the body length ($D > 2B$).
- **shooting_star(d,i)**: Checks for bearish close and long upper shadow.
- **dragonfly_doji(d,i)**: A doji with a disproportionately long lower shadow.

- **gravestone_doji(d,i)**: A doji with a long upper shadow.
- **spinning_top(d,i)**: Small-bodied candle.
- **long_white(d,i)** and **long_black(d,i)**: Large bullish and bearish candles respectively.
- **bullish_marubozu(d,i)** and **bearish_marubozu(d,i)**: Candles with no shadows ($U < \epsilon, D < \epsilon$).
- **bullish_belt_hold(d,i)** and **bearish_belt_hold(d,i)**: Large-bodied candles with minimal shadows at one end.
- **inverted_hammer(d,i)** and **hanging_man(d,i)**: Shadows at opposite ends indicating potential reversal.
- **long_legged_doji(d,i)**: Doji with extended shadows in both directions.
- **rickshaw_man(d,i)**: Doji centered around midpoint of high-low range.
- **high_wave(d,i)**: Small body with long shadows, representing high market volatility.

Each function uses the helper functions to isolate criteria, converting visual rules into boolean logic.

6.4 Two-Candle Pattern Functions

Two-candle functions compare current and previous candles:

- **bullish_engulfing(d,i)**: Confirms small bearish candle followed by larger bullish candle fully engulfing previous body.
- **bearish_engulfing(d,i)**: Reverse scenario.
- **bullish_harami(d,i)** and **bearish_harami(d,i)**: Small second candle inside first body.
- **harami_cross(d,i)**: Second candle is a doji inside the first body.
- **counterattack(d,i)**: Closes near previous candle close after opposing open.
- **matching_high(d,i)**: Highs approximately equal, quantified by GAP_TOL.
- **kicking(d,i)**, **separating_lines(d,i)**, **tasuki_gap_up(d,i)**: Special two-candle patterns involving gaps or continuation signals.

All these patterns use sequential candle properties with thresholds to ensure precise detection.

6.5 Three-, Four-, and Five-Candle Pattern Functions

Longer patterns involve sequences of three to five candles:

- **morning_star(d,i)** and **evening_star(d,i)**: Three-candle reversal patterns using body ratios and midpoint calculations.
- **three_white_soldiers(d,i)** and **three_black_crows(d,i)**: Trend continuation patterns using sequential bullish or bearish closes.
- **three_inside_up/down**, **three_outside_up/down**: Mixed reversal or continuation signals using body ratios.
- **tri_star(d,i)**: Sequence of three dojis.
- **unique_three_river_bottom(d,i)**, **ladder_bottom(d,i)**: Specific reversal patterns with multi-step verification.
- **concealing_baby_swallow(d,i)**, **three_line_strike(d,i)**, **rising_three_methods_ext(d,i)**: Four-candle extended patterns requiring hierarchical checks of individual candle properties.

The logic relies on combining body size, shadow length, directional analysis, and sequence validation.

6.6 Pattern Registry and Detection Engine

FLUEY organizes all pattern functions into a registry:

- Tuple structure: (# of candles, pattern name, function reference)
- Registry ordered by candle count: Longer patterns evaluated first to prioritize complex sequences.

The detection engine iterates through each index of a dataset:

1. For index i , evaluate patterns starting from longest to shortest.
2. Apply pattern function to sequential candle slices.
3. Assign first matching pattern label.

This ensures deterministic labeling while maintaining computational efficiency.

6.7 Algorithmic Considerations

FLUEY achieves scalability and reproducibility due to:

- Vectorized operations through Pandas and NumPy
- Use of helper functions to minimize repeated calculations
- Modular design for easy addition of new patterns
- Thresholds tunable per asset, timeframe, or volatility regime

6.8 Summary

This comprehensive code walkthrough demonstrates how FLUEY translates heuristic visual candlestick patterns into a fully algorithmic, mathematically rigorous system. By combining helper functions, modular pattern definitions, and a hierarchical registry, FLUEY ensures accurate detection of single- through multi-candle patterns across diverse datasets. The system provides a foundation for quantitative research, backtesting, and algorithmic trading strategies.

7 Discussion and Conclusion

7.1 Discussion

The FLUEY framework represents a synthesis of financial technical analysis and rigorous computational modeling, translating qualitative candlestick heuristics into a precise, quantitative system. By systematically defining candle properties such as body size, shadow length, and directional bias, and then combining these properties across sequences of candles, FLUEY creates a deterministic method to identify complex price action patterns. This section elaborates on the broader implications, applications, and limitations of the framework.

7.1.1 Technical Implications

From a technical standpoint, FLUEY demonstrates that patterns traditionally identified visually by traders can be rigorously codified. Each pattern's criteria, once abstract and subjective, now have clear thresholds:

- **Single-Candle Patterns:** By computing normalized body-to-range ratios, shadow lengths, and directional bias, the system captures micro-level market psychology, such as indecision (Doji), potential reversal (Hammer), or trend strength (Marubozu).

- **Multi-Candle Patterns:** By evaluating sequential candles, FLUEY captures dynamics of price evolution, such as reversal points (Morning Star) and continuation trends (Three White Soldiers). These patterns require both intra-candle analysis and inter-candle relationship evaluation, demonstrating a hierarchy of temporal complexity.

This computational approach allows for backtesting, statistical validation, and integration into algorithmic trading, which is difficult when patterns are identified solely visually. By formalizing thresholds like `DOJI_BODY`, `LARGE_BODY`, and `LONG_SHADOW_MULT`, the framework ensures repeatability across datasets, assets, and timeframes.

7.1.2 Practical Implications

Beyond the technical domain, FLUEY provides a framework accessible to everyday market participants:

- **Transparency:** Traders can see precisely why a candle or sequence is classified as a pattern.
- **Education:** FLUEY offers a bridge between visual intuition and quantitative reasoning, helping novices understand market psychology in a rigorous way.
- **Adaptability:** Thresholds are tunable based on volatility, asset class, or trading strategy, making the system relevant to both retail and professional contexts.

By formalizing visual heuristics, FLUEY reduces ambiguity, increases reproducibility, and allows automated systems to interpret price action in a human-aligned manner.

7.1.3 Enhancements over Traditional Methods

Traditional pattern identification suffers from observer bias, inconsistency, and subjective interpretation. FLUEY mitigates these issues by:

1. **Quantitative Standardization:** Every visual rule is expressed mathematically, allowing reproducible detection.
2. **Sequence Analysis:** Multi-candle patterns are analyzed holistically, rather than candle-by-candle, reflecting the true temporal structure.
3. **Integration with Data Pipelines:** Vectorized computation with Pandas and NumPy enables analysis of large datasets in real time.
4. **Modular Extensibility:** New patterns can be incorporated by defining additional functions and adding them to the registry, maintaining scalability.

7.1.4 Limitations

While FLUEY provides a rigorous framework, several limitations are notable:

- **Threshold Sensitivity:** Patterns depend on preset thresholds; improper tuning may lead to false positives or missed signals.
- **Market Context:** Candlestick patterns alone cannot capture macroeconomic conditions, volume, or news-driven events that affect price behavior.
- **Overfitting Risk:** Excessive reliance on historical pattern definitions may fail under novel market regimes.

Awareness of these limitations is crucial for responsible application and interpretation.

7.2 Conclusion

FLUEY bridges the gap between traditional visual technical analysis and modern computational methods. By encoding patterns from single-candle to five-candle sequences with precise mathematical definitions and modular code, FLUEY achieves several key outcomes:

1. Provides deterministic, reproducible pattern detection that can scale to large datasets.
2. Preserves the interpretability of traditional candlestick heuristics while enabling quantitative research.
3. Offers practical utility for both professional traders and everyday market participants seeking systematic tools.

In essence, FLUEY transforms qualitative intuition into quantitative rigor. It is not merely a pattern detection tool but a framework for understanding price behavior at multiple temporal scales. Future extensions may include integration with volume analysis, sentiment data, and machine learning models, further enhancing predictive power and adaptability. Through FLUEY, the principles of technical analysis become not only programmable but also scientifically testable, creating a foundation for both education and research in financial markets.

8 References and Acknowledgments

8.1 References

The FLUEY framework builds upon decades of research in technical analysis, computational finance, and pattern recognition. While traditional candlestick literature provides the foundation for pattern definitions, the formalization into quantitative rules draws from both classical and modern computational methods. The following references were particularly influential in informing the methodology and implementation of FLUEY:

1. Nison, S. (1991). *Japanese Candlestick Charting Techniques: A Contemporary Guide to the Ancient Investment Techniques of the Far East*. New York: New York Institute of Finance. This seminal text provides the foundational definitions for candlestick patterns and their interpretations, serving as the primary source for translating visual heuristics into algorithmic rules.
2. Murphy, J. J. (1999). *Technical Analysis of the Financial Markets*. New York: New York Institute of Finance. Murphy’s comprehensive analysis of charting techniques contextualizes the psychological and behavioral factors underlying candlestick formations.
3. Brock, W., Lakonishok, J., & LeBaron, B. (1992). *Simple Technical Trading Rules and the Stochastic Properties of Stock Returns*. *Journal of Finance*, 47(5), 1731-1764. This work demonstrates the quantitative evaluation of trading rules, supporting the empirical application of FLUEY to historical datasets.
4. Chan, E., & Shelton, T. (2013). *Algorithmic Trading and Quantitative Strategies*. New York: Wiley. This book provides guidance on integrating technical signals into automated trading systems, which informed the registry-based architecture of FLUEY.
5. Murphy, J. J., & Schabacker, R. (2017). *The Art of Candlestick Charting*. This resource helped bridge classical pattern heuristics with the computational formalization adopted in FLUEY.
6. Hochreiter, S., & Schmidhuber, J. (1997). *Long Short-Term Memory*. *Neural Computation*, 9(8), 1735-1780. While focused on deep learning, LSTM architectures influenced the potential integration of sequential pattern recognition with machine learning models in future FLUEY extensions.
7. VanderPlas, J. (2016). *Python Data Science Handbook: Essential Tools for Working with Data*. O’Reilly Media. Provided practical insights for leveraging Pandas and NumPy for high-performance time series computations.

8. Lo, A., & MacKinlay, A. (1999). *A Non-Random Walk Down Wall Street*. Princeton University Press. This reference emphasizes statistical rigor in assessing technical signals, supporting the reproducibility and evaluation of FLUEY outputs.
9. Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction*. MIT Press. Considered for future enhancements involving adaptive thresholding and automated optimization of candlestick-based strategies.
10. Shreve, S. (2004). *Stochastic Calculus for Finance*. Springer. Provided theoretical underpinnings for integrating probabilistic modeling with pattern-based signals.

8.2 Acknowledgments

The development of the FLUEY framework benefited from the combined contributions of computational finance literature, open-source Python libraries, and insights from both professional and retail traders. In particular:

- The Python programming community, including maintainers of Pandas and NumPy, for providing high-performance tools for financial data analysis.
- Researchers in technical analysis, whose rigorous exploration of candlestick patterns provided the basis for the formal definitions implemented in FLUEY.
- Educators and mentors who emphasized the importance of reproducibility, clarity, and computational rigor, inspiring the modular and deterministic design of the pattern registry and detection engine.
- Early beta users and traders who provided feedback on pattern detection accuracy and interpretability, guiding iterative refinement of thresholds and multi-candle pattern logic.

8.3 Closing Remarks

The FLUEY framework not only codifies candlestick heuristics into precise mathematical definitions but also bridges the gap between human intuition and algorithmic rigor. By integrating deterministic functions, sequential evaluation, and modular architecture, FLUEY provides a robust, reproducible, and interpretable system for pattern detection. Its flexibility allows researchers, educators, and traders to adapt, extend, and evaluate patterns across diverse assets and market conditions.

Furthermore, FLUEY serves as a foundation for future exploration:

1. Integration with machine learning models to discover adaptive thresholds or hidden predictive relationships.
2. Expansion to incorporate volume, order flow, and macroeconomic indicators for enhanced signal fidelity.
3. Real-time deployment in algorithmic trading pipelines, demonstrating its scalability and practical utility.
4. Educational applications, enabling learners to understand the quantitative underpinnings of technical analysis.

In conclusion, FLUEY transforms candlestick analysis from a subjective, visual exercise into a precise, computationally tractable methodology. It promotes transparency, reproducibility, and analytical rigor, providing a bridge between classical technical analysis and modern quantitative finance research.